

Beyond Self-Play: Learning Across a Space of Computational Environments

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ABSTRACT

Self-play reinforcement learning typically studies how agents improve within a fixed environment. This research note proposes extending that paradigm by treating the environment itself as a variable in the learning process. The proposed framework constructs families of related computational environments whose rules can be systematically varied, allowing agents to train through self-play across both familiar and previously unseen environments. Using a game-like environment as an initial testbed, the study would investigate whether exposure to structured environmental variation produces strategies that generalize beyond the specific environments encountered during training. The framework also creates an opportunity to study how environmental structure influences learning dynamics, strategy diversity, and transfer. Rather than presenting a completed empirical study, this note outlines the research question, experimental framework, and initial hypotheses that motivate future implementation and experimentation.

Keywords: Self-Play Reinforcement Learning, Environment Generalization, Emergent Strategy

RESEARCH QUESTION

Can self-play be used not to learn strategies within a fixed environment, but to study how strategies emerge, transfer, and change across a systematically varying family of environments?

MOTIVATION

Self-play provides a mechanism through which agents can generate increasingly difficult experiences without requiring a fixed collection of human demonstrations. The traditional formulation, however, generally keeps the environment fixed while the agents improve. I became interested in a different question:

What happens when the environment itself becomes part of the experimental variable?

Rather than training an agent on one immutable game, construct a family of related environments whose rules can be systematically modified. For example, in a Royal Game of Ur-style environment, experiments could vary:

- board geometry,
- number of safe squares,
- number of alley squares,
- rosette locations,
- rosette rules,
- movement rules,
- dice distributions,
- and other structural parameters.

PROPOSED FRAMEWORK

Decompose the training regime into an offensive and defensive coach. Include a narrator of the current game state to train an independent player agent through self-play within an environment (E_θ), where (θ) controls the environment's structure including the opponent strategy. The player agent learns to make a decision using multiple sources of independent information rather than how to play a specific game. The experiment then varies (θ) and measures:

1. learning dynamics,
2. final performance,
3. strategy diversity,
4. transfer to unseen environments,
5. sensitivity to environmental structure.

Reinforcement reward for player action is tied to piece trajectory rather than decision outcome. The central comparison remains fixed between: **fixed-environment learning** and **learning across a distribution of related environments**.

HYPOTHESES

If agents encounter sufficiently diverse environments during training, they may be forced to learn strategies that capture more general structure rather than exploiting idiosyncrasies of one rule set. This can be tested directly through held-out environments.

This shaping of the reward landscape is intended to reduce sparsity and exploit the underlying local structure. The proposed approach takes advantage of natural cycles and simple outcomes to improve reward landscape density. If agents optimize actions over trajectories instead of choice outcomes, the player agent learns trajectory distributions rather than transition probabilities.

Both of the above hypotheses can be tested directly through held-out environments.

EXPERIMENTAL DESIGN

A minimal first experiment would compare:

Training	Test Environment	Question
Fixed (E_1)	(E_1)	Standard self-play
Fixed (E_1)	(E_2)	Transfer
($E_1, \dots, E_k$)	Seen environments	Multi-environment learning
($E_1, \dots, E_k$)	(E_{k+1})	Generalization

Table 1. Proposed minimal experimental design to explore four key questions.

WHY THIS IS INTERESTING

The objective is not to build another game-playing benchmark. The broader question is whether **systematically generated computational environments can become a source of training diversity and experimental discovery**.

This creates a potential bridge between:

environmental design $\leftrightarrow$ self-play $\leftrightarrow$ search $\leftrightarrow$ generalization.

OPEN QUESTIONS

1. Does decomposition of reward information improve model performance?
2. Can agents transfer strategies to environments they have never encountered?
3. Can environment distribution itself be optimized?
4. Can learned strategies reveal structural properties of the environment?
5. Can this framework extend beyond games to abstract reasoning or optimization environments?

Status: Preliminary Research Direction

Code: https://github.com/algorithmic-andy/royal_ur

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